D5 — Ver1 Time-Series Anomaly Detection Framework (Sample Data)

Objective

Develop the first version of the process anomaly detection and KPI variation analysis framework using sample/historical mTorres datasets.

The framework focuses on:

- process variation detection,
- KPI trend analysis,
- process instability identification,
- anomaly indication,
- process health monitoring.

Expected Output

- Time-series monitoring framework
- Process anomaly detection logic
- KPI trend monitoring
- Drift/spike detection
- Process health indicators
- Initial anomaly visualization outputs

Total Estimated Duration

~8 Weeks

Detailed Tasks

D5.1 — Define Time-Series Monitoring & Anomaly Detection Strategy

Duration: ~1 Week

Description

This task focuses on defining how runtime process behavior will be monitored and how anomalies will be detected from process KPIs and machine signals.

The objective is to identify:

- what process variations are important,
- what abnormal behavior means,
- how anomaly severity should be interpreted,
- and which process states require monitoring.

This task establishes the foundation for the entire anomaly framework.

Activities

Study Sample Process Signals

Analyze available sample mTorres datasets to understand:

- signal characteristics,
- KPI evolution patterns,
- process state transitions,
- typical operating ranges.

Example signals:

- tow pressure,
- tension,
- speed,
- temperature,

- compaction force.

Identify Critical KPIs for Monitoring

Identify which KPIs strongly influence:

- process stability,
- defect formation,
- abnormal process conditions.

Examples:

- sudden tow pressure drop,
- unstable tension oscillation,
- abnormal temperature rise.

Define Anomaly Categories

Define different categories of process anomalies such as:

- drift anomalies,
- spike anomalies,
- oscillation anomalies,
- process instability,
- abnormal state transitions.

Define Monitoring Logic

Define:

- continuous monitoring logic,
- moving-window monitoring,
- process-state-based monitoring,
- KPI envelope monitoring.

Define Severity Levels

Define how anomalies are classified:

- warning,
- critical,
- severe instability.

Expected Output

- Time-series monitoring strategy document
- Anomaly classification definition
- KPI monitoring specification

D5.2 — Implement KPI Trend Monitoring Framework

Duration: ~1.5 Weeks

Description

This task focuses on developing the actual KPI monitoring framework used to continuously observe process behavior over time.

The goal is to track:

- KPI evolution,
- process trends,
- operational stability,
- abnormal variations.

This framework becomes the base layer for anomaly detection.

Activities

Implement KPI Trend Tracking

Develop workflows to continuously track KPI changes over time.

Examples:

- tow pressure evolution,
- tension variation,
- process temperature behavior.

Develop Moving Window Analysis

Implement moving-window analysis to evaluate:

- local process behavior,
- temporary instabilities,
- process fluctuations.

Develop Process Envelope Monitoring

Define and monitor:

- acceptable operating ranges,
- normal KPI envelopes,
- warning boundaries.

Implement Temporal KPI Analytics

Compute:

- moving averages,
- variance,
- trend gradients,
- temporal deviations.

Associate KPIs with Process States

Ensure KPI monitoring is aware of:

- layup phase,
- cut/add phase,
- approach,
- retract.

This is important because:

- some KPI variations may be normal in one state,
- but abnormal in another.

Expected Output

- KPI trend monitoring framework
- Temporal KPI analysis module
- Process envelope monitoring module

D5.3 — Implement Drift, Spike & Process Instability Detection

Duration: ~2 Weeks

Description

This task focuses on implementing the core anomaly detection logic responsible for identifying abnormal process behavior.

The objective is to detect:

- gradual process degradation,
- sudden abnormal events,
- unstable process oscillations,
- unexpected process transitions.

Activities

Implement Drift Detection

Detect slow changes in KPI behavior over time.

Examples:

- slowly increasing tension,
- gradual tow pressure instability,

- progressive temperature drift.

Implement Spike Detection

Detect sudden abnormal KPI changes.

Examples:

- sudden pressure drops,
- sharp tension spikes,
- unexpected temperature peaks.

Implement Oscillation Detection

Detect unstable periodic variations which may indicate:

- machine instability,
- process instability,
- vibration-related issues.

Detect Process Instability Patterns

Analyze combined KPI behavior to identify:

- unstable operating regions,
- abnormal process conditions,
- inconsistent process execution.

Associate Anomalies with Process States

Map anomalies to:

- process phases,
- activation states,
- machine operations.

This improves:

- anomaly interpretation,
- HMI visualization,

- process understanding.

Expected Output

- Drift detection module
- Spike detection module
- Instability detection framework

D5.4 — Develop Anomaly Severity & Process Health Framework

Duration: ~1.5 Weeks

Description

This task focuses on translating raw anomaly detections into meaningful process-health information.

Instead of only saying:

Anomaly detected

the framework should estimate:

- severity,
- process risk,
- operational stability.

Activities

Define Process Health Indicators

Develop process health metrics representing:

- process stability,
- operational consistency,

- anomaly density,
- KPI reliability.

Develop Anomaly Scoring Logic

Assign severity scores to anomalies based on:

- duration,
- magnitude,
- affected KPIs,
- process state.

Define Severity Classification

Define categories such as:

- informational,
- warning,
- critical,
- severe process instability.

Develop KPI Stability Indicators

Generate metrics representing:

- stable operation,
- unstable process trends,
- process degradation.

Prepare HMI-Compatible Anomaly Outputs

Prepare anomaly outputs suitable for:

- dashboard visualization,
- anomaly overlays,
- process alerts.

Expected Output

- Anomaly severity framework
- Process health indicators
- HMI-ready anomaly outputs

D5.5 — Validate Time-Series Framework using Sample Data

Duration: ~2 Weeks

Description

This task validates the anomaly detection framework using available sample datasets.

The objective is to verify:

- anomaly accuracy,
- KPI monitoring correctness,
- process-state interpretation,
- process health indicators.

Activities

Execute Monitoring on Historical Datasets

Run anomaly framework on:

- sample process logs,
- historical machine data,
- known abnormal cases.

Validate KPI Trend Interpretation

Verify:

- KPI tracking quality,
- process envelope consistency,
- anomaly triggering logic.

Analyze False Positives & Missed Anomalies

Evaluate:

- false alarms,
- missed detections,
- threshold sensitivity.

Review Results with Process Experts

Validate:

- engineering plausibility,
- process interpretation,
- anomaly meaning.

Refine Detection Parameters

Adjust:

- thresholds,
- anomaly sensitivity,
- process envelope definitions.

Expected Output

- Validated anomaly detection framework
- Validation report
- Refined KPI thresholds
- Updated process health indicators

You have given only for 5 Similarly can you give for all deliverables and tasks

D1 — Requirement Gathering & Validation

Objective

Understand the exact industrial expectations, scope boundaries, KPI requirements, anomaly detection goals, behavioral modeling expectations, and HMI visualization requirements for WP5.3.1.

This deliverable ensures that the process intelligence framework is aligned with:

- OMTC expectations,
- mTorres process understanding,
- WP5.1 interfaces,
- and the revised project scope from latest MoMs.

Expected Output

- Requirement specification document
- KPI & process variable definition
- WP5.1 ↔ WP5.3 responsibility matrix
- Initial use-case definition
- Stakeholder validation report

Total Estimated Duration

~3 Weeks

Detailed Tasks

D1.1 — Review MoMs, Stakeholder Inputs & Existing Process Flow

Duration: ~4 Days

Description

This task focuses on understanding the overall project expectations and identifying the exact role of WP5.3 within the larger process optimization architecture.

The objective is to consolidate:

- MoMs,
- technical discussions,
- stakeholder expectations,
- process optimization goals,
- and process monitoring responsibilities.

This activity is important because the scope has evolved significantly and responsibilities must be clearly separated between WP5.1 and WP5.3.

Activities

Review Existing MoMs & Technical Discussions

Study:

- OMTC discussions,
- mTorres process understanding,
- WP5.1 scope,
- visualization discussions,
- anomaly detection expectations.

Understand AFP/Laying Process Workflow

Understand:

- process phases,
- machine execution flow,

- process activation logic,
- machine state transitions,
- defect occurrence scenarios.

Identify Expected WP5.3 Capabilities

Clarify expectations regarding:

- anomaly detection,
- KPI variation analysis,
- empirical behavioral modeling,
- HMI visualization,
- process intelligence workflows.

Expected Output

- Initial requirement understanding report
- Process optimization expectation summary

D1.2 — Define WP5.1 vs WP5.3 Scope Boundaries

Duration: ~3 Days

Description

This task defines the separation of responsibilities between WP5.1 and WP5.3 to avoid overlap and architectural ambiguity.

The goal is to clearly identify:

- what belongs to runtime monitoring infrastructure,
- what belongs to process intelligence,
- what belongs to data ingestion,
- and what belongs to visualization.

Activities

Define WP5.1 Responsibilities

Clarify WP5.1 ownership for:

- data acquisition,
- preprocessing,
- monitoring infrastructure,
- raw KPI collection,
- database/storage infrastructure.

Define WP5.3 Responsibilities

Clarify WP5.3 ownership for:

- time-series anomaly analysis,
- empirical behavioral modeling,
- process intelligence,
- expected vs actual comparison,
- anomaly visualization.

Define Integration Boundaries

Define:

- data handover mechanisms,
- API expectations,
- synchronization responsibilities,
- HMI integration responsibilities.

Expected Output

- WP5.1/WP5.3 responsibility matrix
- Scope boundary definition

D1.3 — Define KPI & Process Variables

Duration: ~4 Days

Description

This task focuses on identifying which process variables and KPIs are important for:

- anomaly analysis,
- behavioral modeling,
- process understanding,
- and HMI visualization.

Activities

Identify Critical Process Variables

Examples:

- tow pressure,
- tension,
- speed,
- temperature,
- actuator positions,
- process states.

Identify KPI Relationships

Understand:

- which KPIs influence process quality,
- which KPIs influence anomalies,
- which KPIs influence process stability.

Define Anomaly Indicators

Identify:

- process drift indicators,
- instability indicators,
- abnormal transitions,
- process degradation signals.

Expected Output

- KPI mapping sheet
- Process variable definition document

D1.4 — Requirement Validation with Stakeholders

Duration: ~4 Days

Description

This task validates the gathered requirements and architectural understanding with all stakeholders before implementation begins.

Activities

Conduct Requirement Review Sessions

Review:

- KPI definitions,
- anomaly expectations,
- modeling scope,
- visualization expectations.

Validate Use Cases

Validate scenarios such as:

- process instability detection,
- expected vs actual comparison,
- anomaly visualization,
- process activation tracking.

Finalize Scope Understanding

Freeze:

- initial deliverables,
- scope boundaries,
- architecture direction.

Expected Output

- Validated requirement document
- Approved scope definition

D2 — High-Level Architecture Definition

Objective

Define the overall process intelligence architecture for:

- time-series anomaly analysis,
- empirical behavioral modeling,
- HMI integration,
- and validation workflow.

Expected Output

- High-level architecture diagrams
- Data flow definition
- System interaction workflow

- Validation workflow definition

Total Estimated Duration

~3 Weeks

Detailed Tasks

D2.1 — Define Time-Series Analysis Architecture

Duration: ~4 Days

Description

Define the architecture for process anomaly detection and KPI variation analysis.

This architecture focuses on:

- temporal KPI analysis,
- drift detection,
- spike detection,
- process instability analysis.

Activities

Define Monitoring Workflow

Define:

- KPI monitoring sequence,
- process-state-aware analysis,
- rolling-window analysis.

Define Anomaly Detection Pipeline

Define:

- anomaly logic flow,
- threshold framework,
- severity classification flow.

Expected Output

- Time-series architecture workflow

D2.2 — Define Behavioral Modeling Architecture

Duration: ~4 Days

Description

Define the architecture for empirical behavioral modeling and system identification.

The objective is to define how the system learns:

- expected process behavior,
- KPI relationships,
- and process dynamics.

Activities

Define Behavioral Modeling Pipeline

Define:

- input-output structure,
- process-state relationships,
- KPI prediction flow.

Define Expected vs Actual Workflow

Define how:

- predicted process behavior,
- actual runtime behavior
 will be compared.

Expected Output

- Behavioral modeling architecture

D2.3 — Define HMI Integration Workflow

Duration: ~4 Days

Description

Define how process intelligence outputs will be integrated into visualization/HMI systems.

Activities

Define Visualization Layers

Define visualization for:

- KPI trends,
- process activation,
- anomaly overlays,

- process health indicators.

Define 3D Overlay Workflow

Define how:

- process anomalies,
- KPI deviations
  are superimposed onto part geometry.

Expected Output

- HMI integration workflow

D2.4 — Define Validation & Refinement Workflow

Duration: ~3 Days

Description

Define the overall validation loop for:

- anomaly logic refinement,
- behavioral model refinement,
- threshold refinement.

Activities

Define Validation Flow

Define:

- expected vs actual comparison,

- anomaly validation,
- threshold refinement.

Define Model Improvement Workflow

Define:

- model update process,
- refinement iterations,
- process validation logic.

Expected Output

- Validation workflow architecture

D3 — Modeling Dataset Preparation using Sample mTorres Data

Objective

Prepare separate datasets for:

- time-series anomaly detection,
- empirical behavioral modeling.

This is necessary because:

- time-series anomaly analysis
  and
- system identification
  require different dataset structures.

Expected Output

- Time-series-ready datasets
- Behavioral-model-ready datasets
- KPI mapping
- Process-state segmentation

Total Estimated Duration

~6 Weeks

Detailed Tasks

D3.1 — Sample Dataset Collection & Understanding

Duration: ~1 Week

Description

Collect and study sample/historical mTorres datasets to begin early development before actual laying data becomes available.

Activities

Collect Sample Process Data

Collect:

- machine signals,
- KPI logs,

- execution states,
- defect/event examples.

Study Dataset Characteristics

Understand:

- signal structure,
- timestamps,
- sampling rates,
- missing values,
- process phases.

Expected Output

- Sample dataset repository
- Dataset understanding report

D3.2 — Define Inputs/Outputs for Behavioral Modeling

Duration: ~1 Week

Description

Define the process inputs and outputs used for empirical modeling.

Activities

Define Inputs

Examples:

- speed,

- temperature,
- tow pressure,
- machine states.

Define Outputs

Examples:

- process quality,
- process stability,
- KPI trends,
- anomaly indicators.

Expected Output

- Behavioral model IO mapping

D3.3 — Define Process States & Process Activation Logic

Duration: ~4 Days

Description

Define operational process states for:

- process understanding,
- anomaly analysis,
- HMI visualization.

Activities

Define Process States

Examples:

- layup,
- approach,
- retract,
- cut/add,
- transitions.

Define Activation Logic

Define:

- state transitions,
- activation sequences,
- execution flow.

Expected Output

- Process-state model

D3.4 — Prepare Time-Series Analysis Datasets

Duration: ~1.5 Weeks

Description

Prepare datasets specifically for:

- anomaly detection,
- KPI variation analysis,
- drift/spike detection.

Activities

Synchronize Signals

Align:

- timestamps,
- KPI signals,
- process states.

Generate Rolling Windows

Create:

- moving windows,
- temporal sequences,
- process intervals.

Generate Monitoring Features

Prepare:

- moving averages,
- variance,
- gradients,
- drift indicators.

Expected Output

- Time-series-ready datasets

D3.5 — Prepare Behavioral Modeling Datasets

Duration: ~1.5 Weeks

Description

Prepare datasets specifically for:

- empirical behavioral modeling,
- system identification,
- KPI prediction.

Activities

Align Inputs & Outputs

Prepare causal relationships:

Input → Process Response

Generate Modeling Features

Prepare:

- input-output sequences,
- process-state associations,
- temporal embeddings.

Prepare Training Datasets

Create:

- train datasets,
- validation datasets,
- prediction sequences.

Expected Output

- Behavioral-model-ready datasets

D3.6 — Dataset Validation & Documentation

Duration: ~4 Days

Description

Validate the prepared datasets before anomaly modeling and behavioral learning begin.

Activities

Validate Synchronization

Verify:

- signal alignment,
- temporal consistency.

Validate Modeling Compatibility

Verify:

- anomaly framework compatibility,
- SI framework compatibility.

Expected Output

- Dataset validation report

can you D4, D6, D7, D8, D9, D10, D11 and D12

D4 — Data Ingestion Pipeline Design, Development & Testing

Objective

Develop the ingestion framework responsible for receiving process/KPI data from WP5.1 and organizing it for:

- time-series anomaly analysis,
- empirical behavioral modeling,
- HMI visualization.

This deliverable ensures that WP5.3 can work independently once WP5.1 starts providing actual laying data.

Expected Output

- Data ingestion framework
- Runtime data interfaces
- KPI synchronization workflow
- Structured ingestion pipeline
- Integration validation report

Total Estimated Duration

~5 Weeks

Detailed Tasks

D4.1 — Define Data Ingestion Architecture & Interfaces

Duration: ~1 Week

Description

Define how WP5.3 receives:

- KPI streams,
- machine signals,
- execution logs,
- process states,
- contextual information
  from WP5.1.

The objective is to establish a scalable and reusable ingestion architecture.

Activities

Define Ingestion Interfaces

Define:

- APIs,
- file structures,
- message structure,
- synchronization mechanisms.

Define Metadata Structure

Define:

- timestamps,
- process IDs,
- state identifiers,
- KPI naming conventions.

Define Data Flow

Define:

- ingestion flow,

- buffering,
- synchronization,
- dataset routing.

Expected Output

- Ingestion architecture specification
- Interface definition document

D4.2 — Develop Data Ingestion Pipeline

Duration: ~2 Weeks

Description

Develop the actual ingestion pipeline responsible for importing and organizing WP5.1 outputs.

Activities

Implement KPI Ingestion

Develop ingestion logic for:

- KPI signals,
- machine telemetry,
- process states.

Implement Synchronization Logic

Ensure:

- aligned timestamps,
- synchronized KPI streams,
- consistent process-state association.

Implement Dataset Routing

Automatically route ingested data into:

- time-series datasets,
- behavioral modeling datasets,
- HMI visualization datasets.

Expected Output

- Working ingestion framework
- KPI ingestion module
- Synchronization framework

D4.3 — Validate Ingestion Pipeline using Sample/Test Data

Duration: ~1 Week

Description

Validate ingestion logic before actual laying data becomes available.

Activities

Execute Ingestion Tests

Run ingestion using:

- sample datasets,
- simulated KPI streams,
- synthetic anomalies.

Validate Synchronization & Routing

Verify:

- dataset integrity,
- signal alignment,
- process-state mapping.

Expected Output

- Integration validation report
- Dataset routing validation

D4.4 — Finalize Ingestion Documentation

Duration: ~1 Week

Description

Prepare final documentation for the ingestion framework.

Activities

Prepare Architecture Documentation

Document:

- ingestion interfaces,
- synchronization logic,
- dataset routing workflow.

Prepare Integration Guidelines

Document:

- expected WP5.1 inputs,

- metadata requirements,
- synchronization assumptions.

Expected Output

- Final ingestion documentation
- Integration workflow guide

D6 — Ver1 Empirical Behavioral Modeling (Sample Data)

Objective

Develop the first version of the empirical/system identification model using sample mTorres datasets.

The goal is to learn:

- expected process behavior,
- KPI relationships,
- process dynamics,
- and expected process response patterns.

Expected Output

- Initial empirical behavioral model
- KPI prediction framework
- Expected process behavior representation
- Expected vs actual comparison workflow

Total Estimated Duration

~10 Weeks

Detailed Tasks

D6.1 — Define Behavioral Modeling Strategy

Duration: ~1 Week

Description

Define the modeling approach used to represent process behavior.

The objective is to determine:

- which SI/ML methods are suitable,
- how process behavior will be represented,
- and which KPIs are predicted.

Activities

Evaluate Modeling Approaches

Evaluate:

- ARX models,
- state-space models,
- regression models,
- temporal ML models.

Define Prediction Scope

Define:

- predicted KPIs,
- expected process outputs,
- process-state dependency.

Expected Output

- Modeling strategy document

D6.2 — Define Input-Output Relationships

Duration: ~1.5 Weeks

Description

Define causal relationships between:

- machine/process inputs,
- process outputs,
- KPI behavior.

Activities

Identify Process Inputs

Examples:

- speed,
- tension,
- tow pressure,
- machine state.

Identify Expected Outputs

Examples:

- KPI stability,
- process quality,
- anomaly likelihood,
- process health.

Define Process Dependencies

Define:

Input → Process Response

relationships.

Expected Output

- Input-output relationship map

D6.3 — Develop Initial Behavioral Models

Duration: ~3 Weeks

Description

Develop empirical/system identification models using sample datasets.

Activities

Train Initial Models

Train:

- ARX models,
- regression models,
- empirical process models.

Learn KPI Relationships

Learn:

- process behavior,
- temporal KPI evolution,
- process-state influence.

Generate Expected Process Behavior

Generate:

- expected KPI envelopes,
- expected process response,
- nominal process behavior.

Expected Output

- Initial empirical behavioral model

D6.4 — Validate Behavioral Models

Duration: ~2 Weeks

Description

Validate behavioral models against sample process data.

Activities

Compare Predictions vs Actual

Compare:

- predicted KPI behavior,
- actual process response,
- process-state transitions.

Analyze Residuals & Errors

Evaluate:

- prediction quality,
- residual patterns,
- instability regions.

Expected Output

- Model validation report
- Prediction performance metrics

D6.5 — Develop Expected vs Actual Comparison Framework

Duration: ~1.5 Weeks

Description

Develop framework for comparing runtime process behavior against expected behavioral predictions.

Activities

Develop Comparison Logic

Compare:

Expected Process Behavior

vs
Actual Runtime Behavior

Generate Deviation Indicators

Generate:

- KPI deviation metrics,
- process instability indicators,
- anomaly correlation.

Expected Output

- Expected vs actual comparison framework

D6.6 — Refine Behavioral Models

Duration: ~1 Week

Description

Improve:

- prediction stability,
- KPI estimation quality,
- process representation.

Activities

Tune Model Parameters

Refine:

- model coefficients,
- temporal parameters,
- process-state relationships.

Improve Prediction Quality

Reduce:

- prediction errors,
- unstable estimations,
- false behavior interpretation.

Expected Output

- Refined behavioral model

D7 — Ver1 HMI Integration & Analysis

Objective

Integrate:

- anomaly indicators,
- KPI trends,
- process activation,
- behavioral outputs
  into visualization workflows.

Expected Output

- KPI dashboards
- Process activation visualization
- Anomaly overlays
- Expected vs actual visualization

Total Estimated Duration

~6 Weeks

Detailed Tasks

D7.1 — Define Visualization Requirements

Duration: ~1 Week

Description

Define how process intelligence outputs should be visualized inside the HMI environment.

Activities

Define KPI Visualization Requirements

Define:

- KPI trends,
- process health indicators,
- anomaly severity indicators.

Define Operator Visualization Requirements

Define:

- process-state views,
- process activation visualization,
- expected vs actual visualization.

Expected Output

- Visualization requirement document

D7.2 — Develop KPI Trend Visualization

Duration: ~1.5 Weeks

Description

Develop visual workflows for displaying KPI evolution and process health.

Activities

Visualize KPI Trends

Display:

- KPI evolution,
- temporal behavior,
- process envelopes.

Visualize Expected vs Actual Behavior

Display:

- predicted KPI behavior,
- actual KPI behavior,
- process deviations.

Expected Output

- KPI dashboard prototype

D7.3 — Develop Process Activation Visualization

Duration: ~1 Week

Description

Visualize:

- process phases,
- process transitions,
- activation timelines.

Activities

Develop State Visualization

Visualize:

- layup,
- cut/add,
- approach,
- retract.

Develop Timeline Visualization

Display:

- process activation sequence,
- state transitions,
- execution timeline.

Expected Output

- Process activation visualization

D7.4 — Develop Anomaly Visualization & 3D Overlay

Duration: ~1.5 Weeks

Description

Visualize:

- process anomalies,
- KPI severity,
- instability regions
  directly on 3D geometry.

Activities

Develop Anomaly Overlay Workflow

Superimpose:

- anomaly indicators,
- severity regions,
- KPI deviations.

Associate Anomalies with Geometry

Map:

- anomalies to part locations,
- process states to geometry,
- instability regions to execution paths.

Expected Output

- 3D anomaly visualization

D7.5 — Validate HMI Workflow

Duration: ~1 Week

Description

Validate visualization usability and correctness.

Activities

Validate Operator Readability

Verify:

- dashboard usability,
- anomaly visibility,
- KPI readability.

Validate Process Understanding

Verify:

- process-state clarity,
- anomaly interpretation,
- geometry overlays.

Expected Output

- Validated HMI workflow

D8 — Ver2 Data Ingestion & Dataset Preparation using Actual Laying Data

Objective

Adapt the ingestion and dataset preparation framework using actual laying data received from WP5.1.

Expected Output

- Actual laying datasets
- Production-ready datasets
- Validated synchronized KPI datasets

Total Estimated Duration

~5 Weeks

Detailed Tasks

D8.1 — Collect & Organize Actual Laying Data

Duration: ~1 Week

Description

Collect:

- actual runtime KPI data,
- execution logs,
- process states,
- process events
  from WP5.1.

Activities

Organize Production Data

Structure:

- KPI streams,
- execution states,
- defect/event records.

Expected Output

- Actual laying dataset repository

D8.2 — Prepare Time-Series Datasets using Actual Data

Duration: ~1.5 Weeks

Description

Prepare actual-production datasets for:

- anomaly monitoring,
- drift analysis,
- KPI variation analysis.

Activities

Generate Production Time-Series Sequences

Prepare:

- rolling windows,
- synchronized KPI sequences,
- process-state-aware sequences.

Expected Output

- Production time-series dataset

D8.3 — Prepare Behavioral Modeling Datasets using Actual Data

Duration: ~1.5 Weeks

Description

Prepare production-ready datasets for:

- system identification,
- behavioral learning,
- KPI prediction.

Activities

Generate Production Modeling Datasets

Prepare:

- input-output sequences,

- process-state associations,
- prediction datasets.

Expected Output

- Production behavioral-model dataset

D8.4 — Validate Production Dataset Quality

Duration: ~1 Week

Description

Validate production datasets before Ver2 modeling begins.

Activities

Validate Synchronization & Completeness

Verify:

- timestamps,
- KPI alignment,
- process-state consistency.

Expected Output

- Production dataset validation report

D9 — Ver2 Time-Series Anomaly Detection Framework (Actual Laying Data)

Objective

Refine and validate anomaly detection using actual laying data.

Expected Output

- Production anomaly framework
- Refined KPI thresholds
- Validated anomaly indicators

Total Estimated Duration

~6 Weeks

Detailed Tasks

D9.1 — Analyze Production KPI Behavior

Duration: ~1 Week

Study:

- real KPI evolution,

- production instability,
- operational variations.

D9.2 — Refine Drift & Spike Detection

Duration: ~2 Weeks

Adapt anomaly logic using:

- actual production behavior,
- real process variations.

D9.3 — Refine Severity & Threshold Logic

Duration: ~1.5 Weeks

Improve:

- warning thresholds,
- anomaly severity classification,
- process envelopes.

D9.4 — Validate Anomaly Detection using Actual Data

Duration: ~1.5 Weeks

Validate:

- false positives,
- anomaly accuracy,

- production anomaly interpretation.

D10 — Ver2 Empirical Behavioral Modeling (Actual Laying Data)

Objective

Refine empirical behavioral models using actual laying data.

Expected Output

- Production-calibrated behavioral model
- Refined KPI prediction
- Improved expected behavior estimation

Total Estimated Duration

~7 Weeks

Detailed Tasks

D10.1 — Retrain Behavioral Models using Production Data

Duration: ~2 Weeks

Use actual laying data to:

- retrain models,
- recalibrate parameters,
- improve KPI estimation.

D10.2 — Validate Production Behavioral Response

Duration: ~2 Weeks

Validate:

- predicted behavior,
- process-state influence,
- process stability estimation.

D10.3 — Refine Expected vs Actual Comparison

Duration: ~1.5 Weeks

Improve:

- deviation analysis,
- KPI prediction comparison,
- process behavior interpretation.

D10.4 — Finalize Production Behavioral Model

Duration: ~1.5 Weeks

Prepare:

- final model,
- final KPI prediction logic,
- final behavioral representation.

D11 — Ver2 HMI Integration & Analysis (Actual Laying Data)

Objective

Validate and refine visualization workflows using actual production behavior.

Expected Output

- Production-ready dashboards
- Production anomaly visualization
- Final process activation visualization

Total Estimated Duration

~5 Weeks

Detailed Tasks

D11.1 — Integrate Production KPI Visualization

Duration: ~1.5 Weeks

Visualize:

- actual KPI behavior,
- production trends,
- process health.

D11.2 — Validate Production Anomaly Visualization

Duration: ~1.5 Weeks

Validate:

- anomaly overlays,
- severity visualization,
- geometry mapping.

D11.3 — Validate Expected vs Actual Visualization

Duration: ~1 Week

Verify:

- KPI comparison clarity,
- deviation interpretation,
- operator readability.

D11.4 — Finalize Production HMI Workflow

Duration: ~1 Week

Finalize:

- dashboard integration,
- visualization workflow,
- production HMI outputs.

D12 — Final Validation & Integration Testing

Objective

Validate the complete WP5.3.1 framework end-to-end using actual laying data.

Expected Output

- Validated anomaly framework
- Validated behavioral model
- Final integrated HMI workflow
- Demonstration package

Total Estimated Duration

~5 Weeks

Detailed Tasks

D12.1 — Execute End-to-End Validation

Duration: ~2 Weeks

Validate:

- ingestion,
- anomaly detection,
- behavioral modeling,
- HMI integration.

D12.2 — Validate KPI Prediction & Anomaly Accuracy

Duration: ~1 Week

Evaluate:

- prediction quality,
- anomaly correctness,
- process understanding quality.

D12.3 — Validate HMI & Visualization Workflow

Duration: ~1 Week

Verify:

- usability,
- visualization clarity,
- geometry overlays.

D12.4 — Final Demonstration & Documentation

Duration: ~1 Week

Prepare:

- final demo,
- technical documentation,
- stakeholder presentation.